6-3-1: Examining Transceivers and Connector Types

At the end of this episode, I will be able to:

1. Identify and explain transceivers and connector types, given a scenario.

Learner Objective: *Identify and explain transceivers and connector types, given a scenario.*

Description: In this episode, the learner will examine various transceiver and connection considerations. We will explore standards, transceiver form factors such as SFP and QSFP, TIA Category standards and more.

- What are common considerations when using wired cabling?
    - What is the current setup being used (most commonly Ethernet, maybe Fiber channel for storage-area networks)
- With Ethernet being the most common, what are some considerations?
    - On the workstation end:
        - Cables = copper vs. fiber
    - Copper
        - Plenum vs. Non-Plenum Cable -plenum cables are coated with flame-retardant materials and are used in air circulation spaces. In contrast, non-plenum cables are used in non-air circulating spaces and are toxic when burnt, and less fire retardant.
        - Cable type
            - 802.3 - (CAT5 = 100 Mbps, CAT5e -1 Gbps, CAT6 - 1 Gbps, CAT6a - 10 Gbps, CAT7 - 10 Gbps, CAT8 - 40 Gbps)
        - Connectors
            - RJ-45
                - A common connector type used for Ethernet
                - Also known as an 8P8C connector
                - Commonly wired with TIA-568 wiring standard
            - TIA-568A
                - Green and orange wire pairs on pins 1 and 2, 3 and 6, respectively, and blue and brown wire pairs remain unchanged on pins 4 and 5, 7 and 8, respectively, in the RJ-45 connector.
            - TIA-568B
                - Orange wire pair on pins 1 and 2, and green wire pair on pins 3 and 6; blue and brown wire pairs are consistent with TIA-568A, on pins 4 and 5, 7 and 8, respectively, in the RJ-45 connector.
            - RJ-11 Connector
                - Standard interface for voice-grade telephone equipment.
                - Compact, modular design.
                - Usually has four or six positions and contacts.
                - Connects to modules by plugging into standard jacks on devices like telephones, modems, and fax machines.
- What higher bandwidth demands, like data centers?
    - Fiber
        - Ethernet and Fiber Channel
        - Single-mode fiber - allows only one light mode to pass through, enabling longer distance transmission. The core diameter is 9 microns
        - Multimode fiber - allows multiple modes, suitable for shorter, high-bandwidth applications. Core diameter 50 & 62.5 micron
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    - Copper
        - Direct Attach Copper (DAC) - a type of twinaxial cable that features two inner conductors for high-speed, short-range data transmission, often used in data centers.
    - Transceiver - a device that transmits and receives data signals, allowing for the conversion and processing of these signals across various communication mediums.
    - Form Factors -
        - Small Form-factor Pluggable (SFP) - A compact, hot-pluggable transceiver used for telecommunication and data communications applications.
        - Quad Small Form-factor Pluggable (QSFP) - A compact, hot-pluggable transceiver used to carry more bandwidth than a single SFP transceiver, commonly used in data centers.
- What are the different cable connectors in networks?
    - Subscriber Connector (SC) - A push-pull style connector that is widely used in fiber optic communications.
    - Local Connector (LC) - A small form-factor fiber optic connector commonly used in high-density applications due to its small size.
    - Straight Tip (ST) - A fiber optic connector with a twist-locking mechanism used in single and multimode networks.
    - Multi-fiber Push On (MPO) - A high-density fiber optic connector designed for data communications, capable of connecting multiple fibers in a single connector.
    - Registered Jack (RJ)11 - A connector type primarily used for telephone wiring, and also known as an 8P8C connectors
    - F-type - A connector mainly used for coaxial cable and is common in television, satellite, and cable modem applications.
    - Bayonet Neill–Concelman (BNC) - A quick connect/disconnect RF connector used for coaxial cable, especially in video and RF applications.

- Additional Resources:
    - If applicable